Study Script - Indoor Study

Black: script; Blue: comments (not read out); Red: actions

Read the Consent

Before we start, I will read the consent form to you.

You are invited to participate in a research study to explore how our AR system that augments objects of interest in the environment can help enhance your awareness of your surrounding environment. This study will begin with an initial interview on your demographic information, visual conditions, and experience with augmented reality. Then, we will introduce our system, and you will conduct an observation task to evaluate the system. Lastly, we will have an exit interview on your experiences and suggestions for the system. The entire study will last around 2 hours. If you feel tired or frustrated, feel free to tell me and take a break. You can take as many breaks as you want. We will provide a compensation of 25 dollars per hour and compensate your travel expenses for up to 30 dollars.

You will be audio- and video- recorded during the study. Audio and video recordings in their original form will only be accessible to the research team. Anonymized images or videos may be included in publications, academic presentations, and classrooms. The recordings will be retained indefinitely to be used in future research.

Now we will have your oral consent for the study. Is it ok if I start recording video and audio?

1. **Do you have any questions about this research study?**
2. **Do you agree to participate in this study?**
3. **What is your full name? What is the date for today?**
4. **Do you agree to be quoted directly in publications without your name?**

Pre-study interview

First, I will ask about your demographic information and visual condition.

1. How old are you?
2. How would you identify your gender?
3. What is your visual condition?
    a. Are you Legally blind?
    b. What is your Diagnosis?
    c. What is your Visual acuity?
    d. What is your Field of view?
    e. How is your contrast sensitivity?

    f. How is your Color vision?
    g. Are you sensitive to light?
    h. What is your depth perception?
4. How long have you had this visual condition?
    a. Is this condition progressive or stable?
5. Are you using any tools to assist with your visual experience?
6. This study is based on Augmented Reality, or AR. It is a technology that overlays digital information onto the real world to create an interactive experience between digital and physical elements. Do you have any prior experience with AR?
    a. If yes, could you share the experience?

Tutorial

[Set up the tutorial scene: put all kitchen objects on the table]
Now I will show you our system. This system will augment important objects around you to make them more visible. By augmenting these objects, we want to help you quickly understand your surroundings, especially in a complex scene.

The system is based on this device called Hololens. It has a transparent head-mounted display that you can see through. On the display there are two screens. Through the screens the system can show augmentations on top of each object. And since you can still see the real world, the augmentations can interact with real world objects.

[Launch app before handing to participant]
Now, let's try the system. Would you like to put it on? There is a knob at the back, you can spin it to adjust the tightness. There is a strap on the top, you can adjust it to fit your head. Would you like me to help you put on the headset?

Can you see the table through the screen? Make sure you are seeing through the middle of the head-mounted display.
In front of you is a messy kitchen table. Let's start the system to augment important objects on this table.
[Start system]
Are you seeing anything on the screen? Can you describe what you see? Is there a yellow outline around some of the objects?

This is our system. It can recognize objects in front of you, but we only augment "objects of interest" that are important in this environment. Objects that are not that important are not augmented. Also, as we will cover later, this system will augment objects differently based on their importance.

Discuss augmentation design: Let's first talk about augmentation design. The yellow outline is one design, but we also have other designs. We have three design options. Let's go through each of them and discuss your preference.

- Outline: our first design is the one you are seeing right now, It's a bright yellow outline on each object. What do you think of this design? What do you like about it? What do you dislike about it?
    - If dislike outline color: Which color works best for you? How about now?
    - If dislike outline thickness: Is the outline too thick or too thin? How about now?
- Solid: the second design is a solid overlay on each object. What do you think of this design? What do you like about it? What do you dislike about it?
    - If dislike overlay color: Which color works best for you? How about now?
- Icon with pointer: the third design is an icon that floats above each object and has a line connecting to the object. What do you think of this design? What do you like about it? What do you dislike about it?
    - If dislike icon color: Which color works best for you? How about now?
    - If dislike icon size: Is it too big or too small? How about now?

Okay. We've seen and customized all three design options. Now, let's go through them again and pick the one you like the most.
[show all augmentation designs with the best customization]
Among the three options, which one works best for you? Why?

Discuss distinguishing objects: We've discussed different augmentation designs, but so far, we assumed all objects are equally important and we augment them with the same design. However, there could be some objects that are more important than others. For example, there might be certain things that are more important and you want to see right away, versus others that are secondary important, so still important but just good to know. We want to use different designs to distinguish object importance so you can capture the most important objects immediately.

We have different designs to distinguish objects. Let's go through each of them and discuss. Let's suppose the knives and glasses are more important because they are sharp and hard to see.

- Yellow / blue: the first design is to use a bright yellow outline for the more important objects and a blue outline for the others. What do you think of this design? What do you like about it? What do you dislike about it?
    - (If dislike specific colors): which color would you want for higher importance? What about secondary importance? How about now?

- Outline + icon / outline: the second design is to use an outline for all objects. For the more important ones, we add an icon to each of them. What do you think of this design? What do you like about it? What do you dislike about it?
- Solid / outline: the last design is to use a solid overlay for the more important objects, and an outline for the others. What do you think of this design? What do you like about it? What do you dislike about it?

Okay, we've seen and customized three ways to distinguish object importance. Let's go through them again and pick the one you like the most.
[show all ways of distinguishing objects with the best customization]
Among these three designs, which one works best for you? Why?

Thank you! Now, for our next task, you will be observing this messy table. Please wear the system and check out everything on the table. Be sure to know each item is.

Counterbalance with latin square: 3 conditions (not wear the headset (N); with same augmentations (S); with different augmentations (D))

Participant	Condition 1	Condition 2	Condition 3
P1	N	S	D
P2	S	D	N
P3	D	N	S
P4	N	S	D
P5	S	D	N
P6	D	N	S
P7	N	S	D
P8	S	D	N
P9	D	N	S
P10	N	S	D
P11	S	D	N
P12	D	N	S

Instructions for kitchen observation task

Now, we will begin our task. This is an observation task. Imagine yourself going to cook on this kitchen table, so you need to observe and understand where things are. In each trial, you will scan the table as quickly as you can, but make sure that you interpret the scene sufficiently enough so that you can cook in this kitchen. You can walk around the table if you want. Once you feel you are familiar enough with the scene, please let me know. We'll ask you questions and ask you to draw a map of the table. We will do a total of six trials.

Do you have any questions about the task?
Let's first have a tutorial session.

At the start of each session, you will sit on the chair facing the kitchen table and close your eyes while I set up the scene. Once the scene is ready, I will say "you can start" and you can open your eyes to begin observing. You can walk around the table if you want. Once you feel you are familiar enough with the scene, please turn towards me. I'll ask you questions about the table and ask you to draw a map.

Do you have any questions before we start?
Okay, we can get started. Just keep in mind that we are not testing you, we are testing the system itself, so definitely don't feel any pressure about the task or the map. Just try to remember the scene as much as you can, but if you can't remember much, that's completely fine.

For each trial:
Please close your eyes.

Okay, start.
[Start timing]

[1. General Impression]
Can you describe the scene based on your impression?

[2. Mental Map]
Here's a paper with a big box representing this table. Please draw a map of the kitchen table as accurately as possible.
After drawing (in each trial):
Thank you! From 1 to 7 (with 1 being "not confident" and 7 being "very confident") how confident do you feel about the accuracy of this map?

[3. Order]
When you observed the table, which objects caught your attention first? Why did they catch your attention?

[4. Mental Map Development]
For these objects you drew, why did they catch your attention?

[After each session with two trials]:
You have finished two trials [with/without the system]. Now let's discuss your experience:

- From 1 to 7, with 1 being "not effective" and 7 being "very effective": how effective is [using your bare eyes without the system / wearing the system with all objects augmented equally / wearing the system with objects distinguished] in helping you interpret the scene? Why?
- [SceneGlance with same augmentation] How much does the visual augmentations change how you understand this scene?
- [SceneGlance with distinguishing] How much does the different visual augmentations for different object importance change how you understand this scene?
- [NASA-TLX] Next I will give you a set of statements. For each statement you'll give a score from 1 to 7, with 1 being "very low" and 7 being "very high". We will discuss your experience of interpreting the scene [using your bare eyes without the system / wearing the system with all objects augmented equally / wearing the system with objects distinguished]:
    - How **mentally demanding** do you feel interpreting the scene? Why?
    - How **physically demanding** do you feel interpreting the scene? Why?
    - How **successful** were you in interpreting the scene? Why?
    - How **insecure, discouraged, stressed, and annoyed** were you? Why?

Now we can take a quick break and start our next trials.

Exit Questions

Before we finish the study, we have some final questions on your experiences with the system and the visual augmentations.

- You have observed the scene with three conditions: using your bare eyes, using the system with the same augmentations, and using the system with different augmentations for different object importance. Let's compare them.
    - Which condition do you like the most? Why?

    - Which condition do you dislike the most? Why?
- In our design, we try to distinguish objects at two levels. Do you think this is necessary? Why?
    - If so, do you think two levels are enough? If not, what granularity would you expect?
    - How do you want to refine the augmentation design to distinguish different object importances?

Thank you. That is the end of the study.